Linear Functions: Slope Characterization & Applied Modeling

1. Fundamental Characterization of Linear Slope

The slope m of a linear relationship connecting variables x (independent) and y (dependent) is given by:

$$m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

Setting $\mathbf{\Delta x = 1}$ yields:

$$m = \frac{\Delta y}{1} \implies \mathbf{\Delta y = m}$$

The Universal Operational Meaning of Slope:

The slope m represents the exact amount the **dependent variable** (y) changes every time the **independent variable** (x) increases by **one unit**.

- If $m > 0$, the dependent variable increases by m per unit increase in input.
- If $m < 0$, the dependent variable decreases by $|m|$ per unit increase in input.
- **Units of Slope:** Always $\frac{\text{Units of } \mathbf{y}}{\text{Units of } \mathbf{x}}$.

2. Classroom Case Study: Hanford Cancer Mortality Modeling

From the Cancer Mortality laboratory, linear regression connects the radioactive exposure index I to annual cancer mortality rate M (measured in cancer deaths per 20,000 residents per year):

$$\mathbf{M = 11.2I + 112}$$

Slope Interpretation: For each 1-unit increase in the exposure index I, cancer mortality increases by 11.2 deaths per 20,000 residents each year.

Classroom Problem Worked in Full:

A mayor of a town with population $P = 115,000$ enacts remediation policies that successfully decrease the local exposure index by $\mathbf{\Delta I = -1.7}$ units. How many cancer deaths does the model predict will be prevented over the next **5** years?

- **Step 1: Calculate the reduction in mortality rate per 20,000 people:**

$$\Delta M = m \cdot \Delta I = 11.2 \cdot (-1.7) = \mathbf{-19.04} \text{ deaths per 20,000 people per year}$$

- **Step 2: Scale to the town's total population ($P = 115,000$):** Determine the number of 20,000-person population blocks:

$$\text{Blocks} = \frac{115,000}{20,000} = \mathbf{5.75} \text{ blocks}$$

- **Step 3: Compute annual deaths prevented town-wide:**

$$\text{Deaths prevented per year} = 19.04 \times 5.75 = \mathbf{109.48} \text{ deaths/year}$$

- **Step 4: Compute cumulative deaths prevented over a 5-year span:**

$$\text{Total deaths prevented} = 109.48 \text{ deaths/year} \times 5 \text{ years} = \mathbf{547.4} \approx \mathbf{547} \text{ lives saved} \quad \checkmark$$